

CSC 101

THE FIVE GENERATIONS OF COMPUTERS

ULTIMATE MASTER EXAM NOTES
Memory Hacks + Exam Traps Edition

Federal University Oye-Ekiti (FUOYE) | 2025/2026 Session
Compiled for: INDIGENE

#	Generation	Key Theme
1	Overview	Why computers evolved in generations
2	1st Generation	1940–1956 — Vacuum Tubes
3	2nd Generation	1956–1963 — Transistors
4	3rd Generation	1964–1971 — Integrated Circuits
5	4th Generation	1971–present — Microprocessors
6	5th Generation	Present & beyond — AI

OVERVIEW — WHY GENERATIONS EXIST

- The computer evolved from a large-sized simple calculating machine to a smaller, much more powerful machine
- Its evolution to the current state is defined in terms of GENERATIONS
- Each generation is defined by a new technological development, resulting in better, cheaper, smaller, faster and more efficient computers
- There are currently FIVE generations, classified by: the technology used (hardware/software), computing speed, physical appearance, and applications

■ **MEMORY HACK:** The 5 generations track ONE repeating pattern: a new core technology → smaller size → faster speed → cheaper cost. If you remember the core TECH of each generation, everything else (size, speed, cost) follows logically.

■ **TRAP:** Students often mix up which generation introduced WHICH technology. The exam loves asking 'which generation introduced X' — memorize the tech-to-generation pairing below as a strict 1-to-1 match, not a vague timeline.

Quick-Match Table — Tech to Generation (MEMORIZE THIS FIRST)

Generation	Core Technology	Years
1st	Vacuum Tubes	1940–1956
2nd	Transistors	1956–1963
3rd	Integrated Circuits (ICs)	1964–1971
4th	Microprocessor	1971–present
5th	Artificial Intelligence	Present & beyond

■ **MEMORY HACK:** VTIMA = Vacuum, Transistor, IC, Microprocessor, AI — say it like a word: 'V-TIM-A'. This is THE single most-tested list in this topic.

1ST GENERATION COMPUTERS (1940–1956)

Core Facts

- Used VACUUM TUBES for circuitry and MAGNETIC DRUMS for memory
- Were often ENORMOUS — taking up entire rooms
- Relied on MACHINE LANGUAGE (the lowest-level, hardest language — pure binary)
- Very expensive to operate; used a great deal of electricity and generated a lot of heat, which often caused malfunctions
- Examples: UNIVAC and ENIAC

Advantages

- It was the only electronic device of its kind at the time
- First device to hold memory

Disadvantages

- Too bulky — large in size
- Vacuum tubes burned out frequently
- Produced a lot of heat
- Constant maintenance problems

■ **MEMORY HACK: 'VUM'** — Vacuum tubes, Uses a lot of power, Machine language. First gen = the HOTTEST, BIGGEST, and HARDEST to program generation. Picture a room-sized oven that only understands 1s and 0s.

■ **TRAP: Don't confuse UNIVAC/ENIAC (1st gen examples) with later 'small computer' examples** — the exam sometimes lists a later-generation device as a distractor option for a 1st-gen question. If it fits in a room and needs vacuum tubes, it's 1st gen, full stop.

2ND GENERATION COMPUTERS (1956–1963)

Core Facts

- TRANSISTORS replaced vacuum tubes — this is the single defining change of 2nd generation
- Moved from cryptic BINARY machine language to SYMBOLIC language
- High-level programming languages emerged: early COBOL and FORTRAN
- These were the FIRST computers to store their instructions in their own memory

Advantages

- Size reduced considerably (compared to 1st gen)
- Much faster
- Very much more reliable

Disadvantages

- Still overheated quickly
- Still had maintenance problems

■ **MEMORY HACK:** Transistor = 'Tiny Replacement' for the vacuum tube. 2nd gen is the generation of FIRSTS: first symbolic language, first high-level languages (COBOL/FORTRAN), first stored-instruction memory.

■ **TRAP:** A very common trick question: 'Which generation FIRST stored instructions in memory?' Students guess 1st gen because it sounds foundational — the real answer is 2ND gen. Lock this in.

3RD GENERATION COMPUTERS (1964–1971)

Core Facts

- The INTEGRATED CIRCUIT (IC) was the hallmark technology of this generation
- Transistors were miniaturized and placed onto silicon chips, called SEMICONDUCTORS
- Instead of punched cards and printouts, users interacted through KEYBOARDS and MONITORS, interfaced with an OPERATING SYSTEM
- Allowed the device to run many different applications at ONE time (multi-tasking begins here)

Advantages

- ICs are very small in size
- Improved performance
- Production cost cheaper

Disadvantages

- ICs are sophisticated (complex) to manufacture/repair

■ *MEMORY HACK: 3rd gen = 'the KEYBOARD generation.'* Before this, humans fed computers punch cards; from this generation on, humans talk to computers directly via keyboard + monitor + OS.

■ **TRAP: Don't confuse INTEGRATED CIRCUIT (3rd gen) with MICROPROCESSOR (4th gen) — they sound similar but are different scales. An IC is transistors on ONE chip performing one function; a microprocessor puts the ENTIRE CPU on one chip. IC = 3rd gen. Microprocessor = 4th gen.**

4TH GENERATION COMPUTERS (1971–present)

Core Facts

- The MICROPROCESSOR brought the fourth generation — thousands of integrated circuits were built onto a SINGLE silicon chip
- The Intel 4004 chip, developed in 1971, located ALL the components of the computer (CPU, memory, input/output controls) on one chip
- Fourth generation computers also saw the development of GUIs (Graphical User Interfaces), the MOUSE, and HANDHELD devices

■ *MEMORY HACK: 4th gen = 'EVERYTHING ON ONE CHIP.' Intel 4004 (1971) is the single most exam-relevant date+name pair in this whole topic — memorize it as a pair, not separately.*

■ **TRAP:** Students often think 'microprocessor' and 'integrated circuit' are interchangeable terms and pick either for either generation. Remember: EVERY microprocessor contains ICs, but not every IC is a microprocessor. The exam tests this exact distinction.

5TH GENERATION COMPUTERS (present and beyond)

Core Facts

- Based on ARTIFICIAL INTELLIGENCE
- Still in development, though some applications already exist — e.g. VOICE RECOGNITION
- Uses PARALLEL PROCESSING and SUPERCONDUCTORS to help make artificial intelligence a reality
- GOAL: to develop devices that respond to NATURAL LANGUAGE input and are capable of LEARNING and SELF-ORGANIZATION

■ *MEMORY HACK: 5th gen = 'the THINKING generation.' Its whole goal (natural language + learning + self-organization) is basically describing what AI assistants aim to do today.*

■ **TRAP: A common wrong answer: saying 5th generation is defined by 'the internet' or 'smartphones.' It is specifically defined by AI-based technology (parallel processing + superconductors + natural language), not any particular device or gadget.**

RAPID FIRE CHEAT SHEET — ALL 5 GENERATIONS

Full Comparison Table

Gen	Years	Core Tech	Key Weakness
1st	1940–1956	Vacuum tubes, magnetic drums	Huge, hot, expensive, machine language only
2nd	1956–1963	Transistors	Still overheated; maintenance issues
3rd	1964–1971	Integrated Circuits (semiconductors)	ICs sophisticated to make/repair
4th	1971–present	Microprocessor (e.g. Intel 4004)	N/A — current dominant tech
5th	Present & beyond	Artificial Intelligence	Still largely in development

Likely Exam Questions & Quick Answers

Question	Answer
What technology defined the 1st generation of computers?	Vacuum tubes (for circuitry) and magnetic drums (for memory)
Name two examples of first-generation computers.	UNIVAC and ENIAC
What replaced vacuum tubes in the 2nd generation?	Transistors
Which generation first stored instructions in memory?	2nd generation
Name two early high-level languages developed in the 2nd generation.	COBOL and FORTRAN
What was the hallmark technology of the 3rd generation?	The Integrated Circuit (IC)
What are transistors on a silicon chip called?	Semiconductors
How did users interact with 3rd-generation computers?	Through keyboards and monitors, interfaced with an operating system
What brought about the 4th generation of computers?	The microprocessor
What chip, released in 1971, put all computer components on one chip?	The Intel 4004 chip
What two input innovations appeared in the 4th generation?	GUIs (Graphical User Interfaces) and the mouse
What is the 5th generation of computers based on?	Artificial Intelligence (AI)
What two technologies help make AI a reality in the 5th generation?	Parallel processing and superconductors
What is the ultimate goal of 5th-generation computing?	Devices that respond to natural language input and are capable of learning and self-organization

YOU GOT THIS, INDIGENE!

FUOYE Nursing Student, Future Neuropsychiatric Nurse & Web Developer

VTIMA. Intel 4004 = 1971. 2nd gen stored instructions FIRST. Now go pass CSC101!